

Can AI detect AI? : The Arms Race between Generation and Detection

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Abstract—Deepfakes have rapidly evolved from containing obvious visual artifacts to exhibiting high levels of photorealism, posing significant risks to digital trust, security, and information integrity. This project investigates whether modern convolutional neural networks (CNNs) truly learn meaningful facial features for deepfake detection or instead rely on spurious correlations present in training data.

Using the publicly available Kaggle real-fake face dataset, we train and evaluate an InceptionV3-based classifier and reproduce high-accuracy performance comparable to recent literature, achieving over 90% training accuracy and 85% validation accuracy. However, a detailed analysis of precision–recall asymmetries reveals substantial class imbalance, indicating that accuracy alone is an insufficient measure of reliability.

To probe the model’s internal reasoning, I applied two complementary explainability techniques : Grad-CAM and LIME to demonstrate that their explanations often diverge, with both methods frequently highlighting background regions rather than semantically meaningful facial features. This inconsistency suggests that the classifier may be exploiting dataset-specific artifacts rather than robust biometric cues.

I further introduce adversarial training using a generative adversarial network (GAN) and FGSM-based perturbations, observing a reduction in overall accuracy but a marked improvement in class balance and robustness.

The results highlight three key findings: (i) high accuracy does not imply reliable or fair detection, (ii) explainability methods can expose hidden dependencies on non-facial features, and (iii) adversarial training improves robustness at the cost of raw performance. We conclude that while AI systems can detect AI-generated media to a limited extent, deepfake detection remains inherently fragile due to the ongoing adversarial arms race between generators and detectors.

Note: This report was authored entirely by Jaee Ponde. The course was taught by Professor Banerjee, and the project was developed under his mentorship.

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I. INTRODUCTION

In recent years, the rapid advancement of deep learning technologies has led to the emergence of highly realistic synthetic media, commonly known as deepfakes [1]. These AI-generated images and videos, often created using techniques such as Generative Adversarial Networks (GANs), can convincingly mimic real human faces, expressions, and speech. While the underlying technology holds immense potential in areas like entertainment, accessibility, and education, its misuse particularly in spreading misinformation, impersonation, and defamation has raised significant ethical and security concerns.

Detecting deepfakes has become an increasingly difficult task. Early iterations often displayed obvious visual artifacts or inconsistencies that made them relatively easy to identify. However, as generation techniques have improved, deepfakes have become more photorealistic and temporally coherent, significantly narrowing the gap between fake and authentic media. This poses a growing challenge for researchers and technologists aiming to build reliable safeguards.

The Analysis of Neural Network Models to Distinguish AI generated faces from Real faces

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Fig. 1. 2024 Paper claiming 100% accuracy

This project was primarily inspired by the paper “Fake vs Real Faces: Deepfake Detection using VGG16 Model” [2]. The authors claim exceptionally high accuracy in detecting deepfakes using a relatively straightforward VGG16-based convolutional neural network. While impressive on the surface, several aspects of the paper raise important questions regarding the robustness and generalizability of the proposed model.

Specifically, the paper makes specious claims about the model’s performance, showcasing near-perfect classification accuracy without providing extensive analysis of potential confounding variables. There is limited discussion around dataset diversity, overfitting risks, or whether the model is learning meaningful facial features versus relying on brittle correlations — such as artifacts in the background, compression noise, or dataset-specific cues. These concerns sparked the central motivation for this project: **to critically evaluate whether such high-performing CNN-based detectors are genuinely focusing on the face itself, or if they are relying on peripheral information like background pixels or image peculiarities.** By reproducing and modifying the architecture, we aim to test the robustness of deepfake detectors under controlled perturbations, thereby assessing how “deep” the detection truly is.

Another key question addressed in this project concerns the explainability of deepfake detection models. As the demand for transparent and trustworthy AI systems continues to grow, it is increasingly important to understand the rationale behind the decisions made by such detectors. In recent years, it has become almost standard practice to attach an explainability module to any trained model, often as a post-hoc justification of its predictions [3]. However, the mere presence of an explanation does not guarantee that it is faithful, reliable, or even meaningful. **This raises a deeper methodological concern: how should one choose between competing explainability methods, and how can we determine which explanations genuinely reflect a model’s decision process rather than offering a convenient but misleading narrative?**

Without principled evaluation criteria, explainability risks becoming a cosmetic addition rather than a tool for building trustworthy systems [4]. Beyond predictive performance, this work investigates whether different explainability techniques provide consistent and reliable evidence for a model’s predictions, or whether they reveal fundamentally different—and potentially conflicting—interpretations of the same decision process.

All code, requirements and cell outputs can be found in my [GitHub Repository](#).

A. Research Questions

Building on the concerns raised about the reliability and robustness of current deepfake detection models, this project is guided by the following key research questions:

1) **Can I train a CNN to achieve very high accuracy on the same dataset?**

The original paper claims near-perfect performance using VGG16. This raises the question of whether such results are reproducible, and more importantly, whether they reflect genuine learning or overfitting to dataset-specific artifacts.

2. Do different explainability models provide consistent explanations? If the CNN truly learns to distinguish between real and fake faces, interpretability tools (like Grad-CAM, LIME, or SHAP) should highlight meaningful facial regions. Inconsistencies across these models may indicate instability or shallow learning.

3) **Can I train a generative adversarial network to do better?**

While the arms race is between detection and generation, I wonder how including an adversary in the training procedure alters our results. It will be of note if the model manages a higher accuracy with adversarial training, indicating better feature understanding. If it deteriorates completely, it exposes the fragile nature of the model’s feature representation.

4) **Can AI detect AI?**

At the core of this project is a meta-question: are machines capable of reliably detecting synthetic content generated by other machines? As generation and detection models evolve together, can detection methods stay ahead—or even keep up?

II. DATA SELECTION & PRE PROCESSING

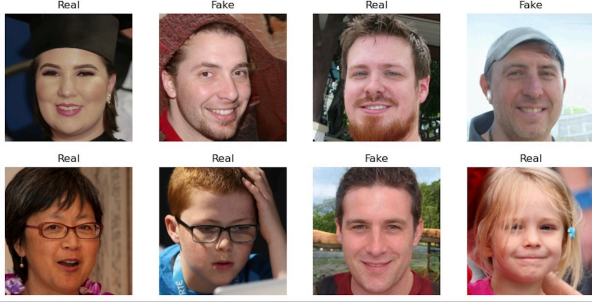


Fig. 2. Snapshot of the Dataset

For this project, I used the publicly available 140K Real and Fake Faces dataset from Kaggle, which was also employed in the study I aimed to reproduce. The dataset contains 70,000 real and 70,000 synthetically generated images, each labeled accordingly.

To ensure a manageable and balanced training setup, I randomly sampled:

- 3,000 images for training (1,500 real, 1,500 fake),
- 1,000 images for validation (500 real, 500 fake), and
- 1,000 images for testing (500 real, 500 fake).

All images were resized and normalized as part of the preprocessing pipeline. For the training set, I applied standard data augmentation techniques to increase the model's robustness and prevent overfitting. The augmentation strategy included:

```
train_datagen = ImageDataGenerator(
    rescale=1./255,          # Normalize
                             # pixel values to [0, 1]
    shear_range=0.2,         # Apply random
                             # shear transformations
    zoom_range=0.2,          # Randomly
                             # zoom in on images
    horizontal_flip=True     # Randomly
                             # flip images horizontally
)
```

The validation and test datasets were only rescaled (without augmentation) to ensure consistent evaluation.

This preprocessing setup mirrors typical practices for image classification with CNNs, while enabling a fair and reproducible comparison with the original paper's results.

III. REPRODUCING A HIGH-ACCURACY CNN FOR ANALYSIS

A. Model Selection

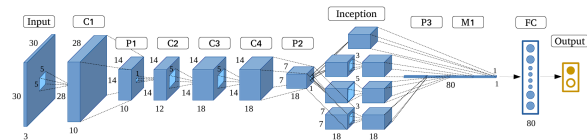


Fig. 3. Inception V3

For the primary detection model, I selected **InceptionV3**, a deep convolutional neural network known for its strong performance on image classification tasks and its efficient architectural design. InceptionV3 uses factorized convolutions, auxiliary classifiers, and aggressive dimensionality reduction, making it both computationally efficient and effective at capturing complex spatial hierarchies in images.

Compared to architectures like VGG16, InceptionV3 also trains much faster making it the ideal choice.

B. Training

The InceptionV3 model was fine-tuned on the selected dataset using the following training configuration:

- **Loss Function: Categorical Crossentropy**

Used due to the binary classification setup (real vs fake), with the data encoded in a one-hot format.

- **Optimizer: Adam Optimizer**

Chosen for its adaptive learning rate and fast convergence properties, particularly useful for training deep networks.

- **Learning Rate: 1e-4**

A small learning rate was used to ensure stable convergence and prevent overshooting during fine-tuning.

- **Epochs: 10**

The model was trained for 10 epochs, which was sufficient to observe convergence while avoiding overfitting on the relatively small subset of the full dataset.

- **Batch Size: 32**

The training and validation losses were monitored to evaluate the model's learning behavior and guide model checkpointing.

C. Results

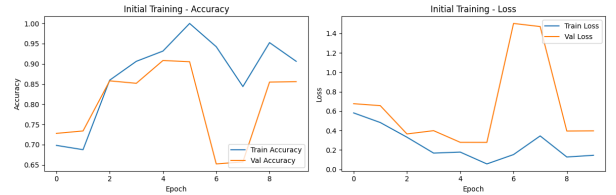


Fig. 4. Training Metrics

The InceptionV3 model achieved the following performance on the training and validation sets after 10 epochs:

- Final Training Accuracy: **90.62%**
- Final Validation Accuracy: **85.58%**

While the model reached high accuracy on both sets, the gap between training and validation accuracy (5%) suggests a degree of overfitting, possibly due to the limited dataset size and the high capacity of InceptionV3. This is why I trained it a few epochs further than the minimum training loss, to ensure we don't pick the most 'overfitted' version. This somewhat resembles the double descent phenomena observed when model parameters far exceed the training data [5]. Nevertheless, the model shows strong discriminative ability and demonstrates that deep CNNs can learn meaningful patterns from synthetic and real facial images.

D. A deeper look

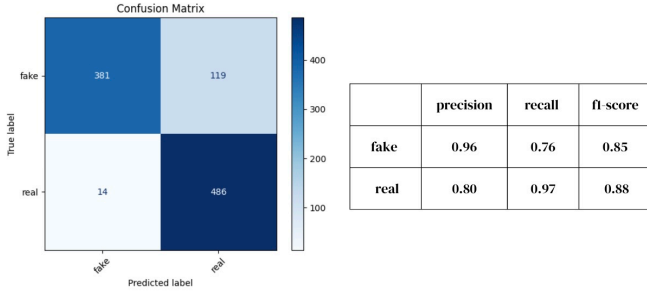


Fig. 5. Inception V3 Test Metrics

The gap between precision and recall for each class is not just critical:

- **For the fake class, the model has very high precision (0.96) but relatively low recall (0.76). This means:**
 - ▶ When the model predicts an image is fake, it is usually correct.
 - ▶ However, it misses many fake images (i.e., false negatives).
 - ▶ The model is conservative when labeling something as fake.
- **For the real class, the model has very high recall (0.97) but lower precision (0.80), which means:**
 - ▶ The model correctly identifies almost all real images.
 - ▶ But it incorrectly labels some fake images as real (i.e., false positives).
 - ▶ The model is more lenient when labeling something as real.

This suggests the model is **skewed towards avoiding false positives for fake images, potentially at the cost of missing actual fakes**. In other words, it tends to err on the side of trusting images to be real unless it's very confident they are fake.

This suggests a large gap in even the existing CNNs built for classification - accuracy is not a measure of skew, balance or robustness. **The model may be 90% accurate, but with alarming imbalance in precision and recall of classes.**

IV. GENERATIVE ADVERSARIAL NEURAL NETWORK

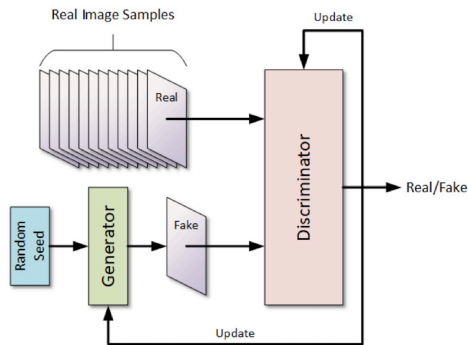


Fig. 6. GAN

A Generative Adversarial Network (GAN) is a type of neural network architecture used in machine learning to generate new data samples that resemble a given dataset. It was introduced by Ian Goodfellow and his colleagues in 2014 and has become a foundational technique in the field of generative modeling.

A GAN consists of two neural networks that compete against each other:

- **Generator (G):**
 - Takes in random noise and tries to produce realistic-looking data (e.g., images, audio).
 - Its goal is to fool the Discriminator into thinking its fake outputs are real.
- **Discriminator (D):**
 - Takes in real and fake data and tries to distinguish between the two.
 - It learns to say: “this is real” or “this is generated.”
- These two networks are trained simultaneously in a game-like setting:
 - The generator improves by learning how to create better fakes.
 - The discriminator improves by becoming better at spotting fakes.
 - Eventually, if the training goes well, the generator becomes so good that the discriminator can't reliably tell real from fake — that's when the GAN is considered successful.

A. Training

As part of evaluating and strengthening the GAN model trained for the real vs fake face classification task, I implemented **adversarial attacks using FGSM (Fast Gradient Sign Method)**. FGSM is a simple yet powerful technique to generate adversarial examples—inputs that have been intentionally perturbed to fool a model.

The process works as follows:

- 1) **Gradient Computation:** For a given input image, I compute the **gradient of the loss with respect to the input pixels**. This tells us how small changes in the image would affect the model's prediction.
- 2) **Adversarial Perturbation:** I then use the **sign of the gradient** to determine the direction in which to slightly modify each pixel. This creates a new image that looks visually similar to the original, but is subtly altered in a way that can mislead the model.
- 3) **Batch-Level Perturbation:** During training or evaluation, I apply this adversarial noise to **30 images per batch**, resulting in a total of around **1000 adversarial examples** across all batches.

B. Results

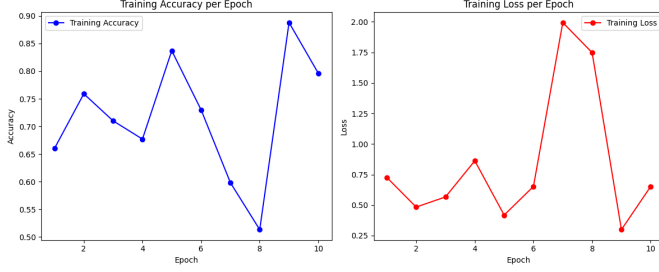


Fig. 7. Results of GAN training

We can see that the accuracy has significantly dropped, to **80%**. But let's take a closer look.

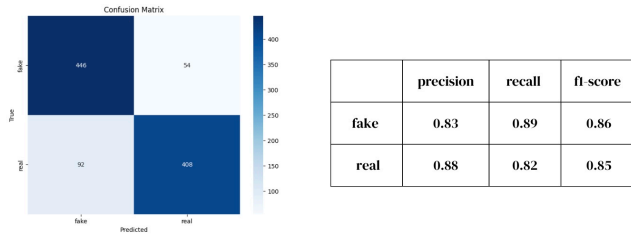


Fig. 8. GAN confusion matrix

C. Comparing with the model trained without an adversary

Inception V3

	precision	recall	f1-score
fake	0.96	0.76	0.85
real	0.80	0.97	0.88

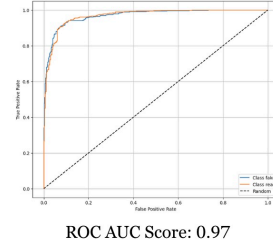
GAN

	precision	recall	f1-score
fake	0.83	0.89	0.86
real	0.88	0.82	0.85

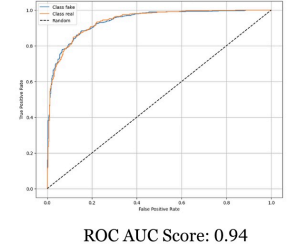
One of the most striking observations from evaluating the GAN-enhanced model is the improved balance in the confusion matrix compared to the non-GAN baseline. In the non-GAN model, there was a clear skew — for example, it might have had a strong bias toward predicting “real” images correctly while consistently misclassifying “fake” ones, or vice versa. This resulted in high precision for one class and low recall for the other, indicating the model was disproportionately confident in one label and underperforming on the other.

In contrast, the GAN-trained model yields a far more symmetric confusion matrix. The numbers of true positives, true negatives, false positives, and false negatives are more evenly distributed across both classes. **This translates to precision and recall values that are roughly equal for both “real” and “fake” classes, demonstrating that the model is not only more fair in its predictions but also more robust in handling subtle differences between the two classes.**

Inception V3



GAN



You can also observe that the ROC curve is not shifted towards a specific class, and is equally close to both the x and y axis, giving an optimal point that is **less accurate, but more balanced**.

V. DO DIFFERENT EXPLAINABILITY MODELS PROVIDE CONSISTENT EXPLANATIONS?

A. GradCam

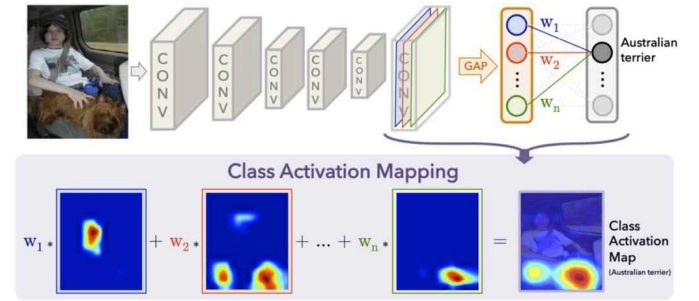


Fig. 9. GradCam

To better understand what features the model was focusing on when classifying images as real or fake, I used Grad-CAM (Gradient-weighted Class Activation Mapping), a widely-used explainability technique for convolutional neural networks [6].

Grad-CAM works by **leveraging the gradients of the predicted class flowing into the final convolutional layer of the model**. These gradients are used to compute the importance of each neuron in that layer with respect to the prediction.

By weighting the feature maps with these importance scores and applying a ReLU activation, Grad-CAM produces a heatmap that highlights the most influential regions in the input image.

This heatmap is then overlaid on the original image to **visualize where the model “looked” when making its decision**.

Unlike black-box predictions, Grad-CAM enables us to inspect whether the model is focusing on meaningful facial features such as the eyes, mouth, or skin texture — or whether it is relying on unrelated background artifacts, lighting inconsistencies, or dataset noise. This interpretability is especially crucial in the context of deepfake detection, where models may achieve high accuracy for the wrong reasons.

For inception v3 I chose the following layers :

```
layer_names = [
    "mixed3",
    "conv2d_1", // Starting Layers
    "conv2d_52",
```

```
"mixed6", // Middle Layers
"conv2d_93",
"mixed10" // Final Layers]
```

a) Results:
Fake Images:



Real Images:



In analyzing the Grad-CAM visualizations for fake images, we observe that the model tends to focus heavily on the background regions across almost all layers. Unlike the clear, feature-focused activations seen in real images, the activations here often lack a coherent or semantically meaningful pattern.

This observation **highlights a common misconception about Grad-CAM**: it's often portrayed as a tool that will neatly outline semantically relevant features, such as eyes, nose, or facial contours. However, this assumption is both misleading and incorrect. Grad-CAM reflects the gradient-based importance of regions relative to the model's prediction, not necessarily human-interpretable semantics.

A given layer's activation map may not visually correspond to meaningful features, and yet, that layer may still play a crucial role in the model's overall decision-making. Semantic representations often emerge progressively and in aggregate across layers, rather than within any single visualization. Therefore, expecting each Grad-CAM output to offer clear semantic insight can result in misinterpretation of the model's actual behavior.

B. LIME

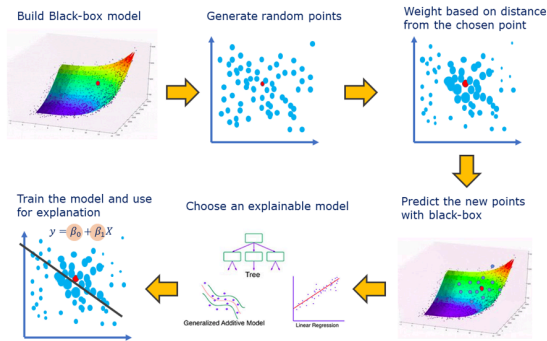


Fig. 10. LIME

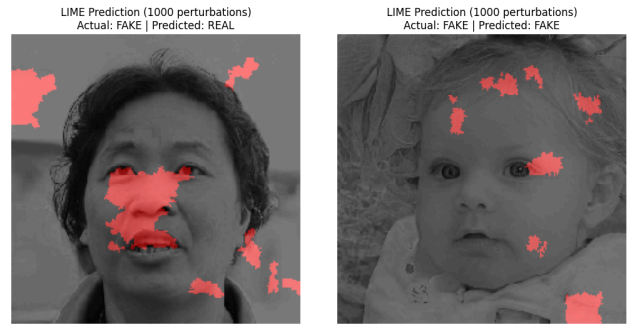
LIME (Local Interpretable Model-agnostic Explanations) is a model-agnostic method, meaning it can be applied to any classifier, not just neural networks. **It works by perturbing the input image — for example, by turning off (masking) different patches — and observing how these changes affect the model's predictions.**

Based on the model's responses to these modified inputs, **LIME fits a simple, interpretable surrogate model (like a linear model) locally around the instance being explained.** This allows it to estimate which regions of the image were most influential in driving the prediction [7]. The result is a visual explanation that highlights the superpixels (grouped regions of similar pixels) that contributed the most to the classification.

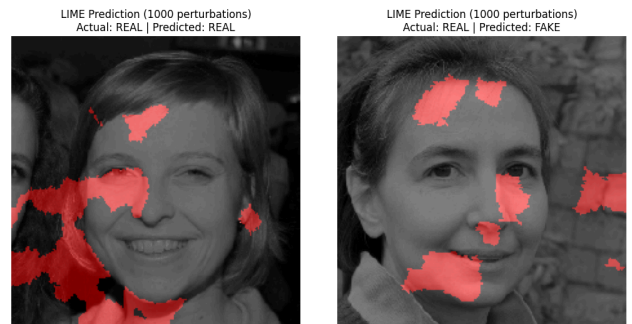
By applying LIME to both real and fake images, I could assess whether the CNN's predictions were driven by meaningful facial features or spurious correlations in the background. Since LIME treats the model as a black box, **it complements Grad-CAM by offering an entirely different interpretability lens — one that focuses on input perturbation rather than internal gradients.**

For the purposes of this study, I used LIME for **1000 perturbations**.

a) Results:
Fake Images:



Real Images:



What's interesting to note about LIME is that it highlights not only the face but also significant portions of the background. **This indicates that the locally trained surrogate model, which approximates the behavior of the original model in the vicinity of a particular input, does rely on the background to make its predictions.** This reliance on background features suggests that the original model might be picking up on spurious correlations — patterns in the data that are not semantically meaningful but are consistently associated with certain classes.

In the context of real vs fake face classification, **this could mean the model has inadvertently learned to associate background textures or lighting artifacts with the “fake” class, which undermines the reliability of its predictions.** LIME, by isolating the most influential image segments regardless of semantic meaning, helps us uncover these biases and emphasizes the importance of evaluating model interpretability tools critically, especially when trust and fairness are essential.

C. Are explanations consistent?



The discrepancy between LIME and Grad-CAM highlights a critical issue in model interpretability: different explainability tools can reveal fundamentally different aspects of a model’s decision-making process.

When their outputs don’t align — for instance, when LIME emphasizes both the face and background while Grad-CAM focuses predominantly on background regions — **it raises a red flag.**

- This suggests that the model might be **relying on irrelevant or misleading features**, such as background textures or lighting artifacts, rather than the actual object of interest. **If we were to strip each image off its background, would it even work?**
- Such behavior points to a deeper problem: the model’s internal representations might be **highly complex yet brittle**, tuned to correlations in the training data that don’t generalize well to real-world examples.
- **It becomes clear that a single interpretability method is not enough.** To meaningfully understand and trust a model, especially in high-stakes domains, we must triangulate insights from multiple tools and critically examine where and why they diverge.

VI. DISCUSSION

This project gave me one of the most important insights: **accuracy alone is not a sufficient measure of a model’s robustness or trustworthiness.**

A Convolutional Neural Network (CNN), for instance, can be trained to achieve **high overall classification accuracy**, but still exhibit **imbalanced performance across classes**. This imbalance is often reflected in skewed **precision and**

recall—where the model might correctly identify one class (e.g., “real”) at the expense of misclassifying the other (e.g., “fake”). Despite a seemingly impressive accuracy score, the model could be failing to generalize fairly across the dataset, making it unreliable for real-world deployment.

Moreover, when applying **model explanation methods like Grad-CAM and LIME**, we often observe **inconsistent or contradictory interpretations**. These tools may highlight different regions of the image as important for the same prediction, indicating that the model’s decision-making process is neither transparent nor easily interpretable. This inconsistency further undermines the confidence in what high accuracy may suggest.

Crucially, **adversarial training**, such as using GANs or applying perturbations through methods like FGSM, **doesn’t always result in a higher validation accuracy.** However, it often **improves the model’s robustness**—its ability to maintain performance in the face of small, targeted distortions or adversarial inputs. A model that withstands adversarial attacks is less likely to be overfitting superficial patterns and more likely to have learned meaningful, generalizable features.

All of this underscores the point that **accuracy is just one dimension of model evaluation.** A truly reliable model must also demonstrate **robustness (resistance to adversarial or distributional shifts)** and **interpretability (clear, consistent reasoning behind decisions)**. Especially in critical applications—like deepfake detection, medical imaging, or security—models must be judged on their ability to generalize, explain, and withstand manipulation, not just on their ability to correctly classify a held-out test set.

VII. CAN AI DETECT AI?

Ultimately, the question “**Can AI detect AI?**” touches on a deeper issue about the limits of adversarial dynamics—and the answer is **not definitively yes.** While AI systems can be trained to detect synthetic content generated by other models, this detection is inherently **fragile and temporary.** Every time we build a network to detect fake images, a more advanced **adversarial network can be trained to bypass that detector.** This cat-and-mouse game can repeat indefinitely—each detector learning current patterns, and each generator learning how to exploit its blind spots.

What makes this even more complex is that the **features used to detect “fakeness” are not fixed.** As generative models improve, their outputs become increasingly indistinguishable from real data—not only to humans but also to machines. Add to this the fact that explainability methods are often unreliable, and adversarial training is reactive rather than preventative, and you begin to see the inherent asymmetry: **generators only need to fool the model once**, whereas **detectors must defend against all possible manipulations.**

So while AI can detect AI to a certain extent, that detection is **always a step behind**, and never absolute. The game is infinite—and for every advance in detection, an equally powerful advance in deception can follow.

In that sense, AI cannot truly and conclusively detect AI, because the very arms race between generation and detection is endless by design.

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